

Dark-sector characterization within TBRME: what the framework requires, constrains, and touches (companion note — constraints and taxonomy, never identity)

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Identity fence (binding). This note does *not* and *cannot* determine what dark matter *is*. TBRME takes the matter QFT as an input to $\langle T_{\mu\nu} \rangle$ and is agnostic about dark-sector field content by construction. Every statement below is one of three kinds: what the framework **REQUIRES** (a category statement), **CONSTRAINS** (its own anchor accounting run against candidate classes), or **TOUCHESES** (which theorems apply to which candidate classes). No sentence identifies, explains, or predicts a dark-matter field content.

Abstract

Within TBRME, gravitational dynamics is Einstein gravity plus a pinned, exhaustively catalogued set of deviations, all either cosmologically homogeneous or decoherence-gated. We record three findings. **(Lane 1, REQUIRES.)** No framework deviation carries a quasi-static galactic footprint; despite the numerical coincidence $a_0 \approx cH_0/2\pi$ the effective- Λ running delivers only $a_\Lambda(r) \approx H_0^2 \Omega_\Lambda r$, smaller than a_0 by $\sim 10^{-5}$ at galactic radii ($a_\Lambda/a_0 \approx 1.2 \times 10^{-5}$ at 10 kpc) and of the wrong sign and radial profile, so rotation-curve/lensing/cluster phenomenology requires genuine stress-energy $\langle T_{\mu\nu} \rangle$. **(Lane 2, CONSTRAINS.)** *Corrected and re-derived (run 112).* The previously quoted $\text{str}[k_1] \approx -14.83$ (whence $g \approx 0.42$) was the statistics-blind plain trace; the signed bosons-minus-fermions supertrace this note names evaluates to the near-cancellation $\text{str}[k_1]_{\text{SM}} = +\frac{1}{6} - 4\xi_H$ ($+\frac{2}{3} - 4\xi_H$ with right-handed neutrinos), sign-indefinite in ξ_H ; the parent now consumes the anchor only as a *posited* $\mathcal{O}(1)$ input. The accounting inverts: the anchor still excludes no standard candidate, but no longer by insensitivity — every individual sub-Planckian candidate shifts $\text{str}[k_1]$ by 100–400% of its whole corrected value (a single massless dark vector is 400%, and nulls it exactly on the right-handed-neutrino variant), so the anchor is co-determined by the dark sector rather than a constraint on it; a hidden sector’s aggregate aligned weight is banded conditionally on the posited anchor, no longer excluded. **(Lane 3, TOUCHES.)** A taxonomy of which framework theorems apply to which candidate class.

1 The deviation budget (lane-1 foundation)

TBRME reproduces Newtonian and post-Newtonian gravity exactly ($\gamma = \beta = 1$; “at Newtonian order there is no deviation from textbook GR”). Its complete set of departures from Einstein gravity is:

	Deviation	Scale	Grade (parent)
D1	H^2 running of Λ_{eff} , ν unobservably small	cosmological (homogeneous)	Proposition
D2	dissipative Clausius- σ , $w \geq -1$, amp. $\sim 10^{-122}$	cosmological (dS horizon)	mechanism-settled
D3	CC variance $\delta\Lambda/\Lambda \sim 10^{-61}$	cosmological (homogeneous)	Proposition
D4	modified-TDSE $H_{\text{eff}} = H_s - \frac{\mu}{2} H_s^2$	mesoscopic (BMV) only	Proposition

D1–D3 enter the effective cosmological constant / Friedmann background and are spatially homogeneous; D4 is unitary, “does not modify the static potential,” and is decoherence-gated off for any

classical (astrophysical) source. Hence none is a position-dependent galactic potential. (Budget exhaustiveness is certified by the m1 arm; the galactic conclusion is robust even against an incomplete cosmological-sector census, since any missing deviation cannot live in the decohered quasi-static regime by the Newtonian-correspondence result.)

2 Lane 1: no gravitational mimic of dark matter (REQUIRES)

The MOND scale $a_0 \approx 1.2 \times 10^{-10} \text{ m/s}^2$ satisfies $a_0/(cH_0) = 0.17\text{--}0.18 \approx 1/2\pi = 0.159$ (verified): the famous coincidence is real. It is not dynamically realized. The only quasi-static footprint of the H^2 term is the Schwarzschild–de Sitter tidal acceleration $a_\Lambda(r) = (\Lambda_{\text{eff}} c^2/3) r = H_0^2 \Omega_\Lambda r$:

r	10 kpc	100 kpc	1 Mpc
$a_\Lambda \text{ (m/s}^2\text{)}$	1.5×10^{-15}	1.5×10^{-14}	1.5×10^{-13}
a_0/a_Λ	$\sim 8 \times 10^4$	$\sim 8 \times 10^3$	$\sim 8 \times 10^2$

It reaches a_0 only at $r \sim 700\text{--}815$ Mpc. Triple disqualification: it is $\sim 10^4\text{--}10^5\times$ too small at galactic radii; it is *repulsive* (outward, $\Lambda > 0$) where a halo needs inward attraction; and it grows as $+r$ where a flat rotation curve needs $a \propto 1/r$.

Proposition 1 (No galactic-scale gravitational mimic of dark matter; *PLAUSIBLE*). *In the decohered, static, weak-field regime governing galactic dynamics, TBRME reduces to ordinary linearised Einstein gravity sourced by the classical $\langle T_{\mu\nu} \rangle$; its distributional/two-boundary apparatus and its unitary modified-TDSE correction leave the static force law unmodified. The only genuine gravitational deviations (D1–D3) are cosmologically homogeneous, and (despite $a_0 \approx cH_0/2\pi$) deliver at galactic radius only $a_\Lambda(r) \approx H_0^2 \Omega_\Lambda r$ — of the wrong magnitude, sign, and radial profile to source a rotation-curve or lensing halo. Consequently no framework deviation can masquerade as dark matter at galactic scales, and any rotation-curve, lensing, or cluster phenomenology within TBRME requires genuine stress-energy $\langle T_{\mu\nu} \rangle$ in the source.*

Grade. PLAUSIBLE: a proof from in-print inputs with a THEOREM-grade core (the Newtonian-correspondence result plus verified arithmetic), conditional on (R1) budget exhaustiveness and (R2) the composite-body μ -caveat, both flagged unresolved in the parent and both orthogonal to the static force law. This is a REQUIRES statement: it makes no claim about the dark sector’s field content.

3 Lane 2: the supertrace / anchor accounting (CONSTRAINS)

Correction (run 112; supersedes the anchor numbers previously printed here). Earlier versions of this note and of the parent’s H*2 set $g := (E_\star/E_{\text{Pl}})^2 \approx 0.42$ ($E_\star \approx 0.65 E_{\text{Pl}}$) from a quoted $\text{str}[k_1] \approx -14.83$. That number is not the value of the printed definition: $-14.83 = -\frac{89}{6} = 4(\frac{1}{6}) + 12(-\frac{2}{3}) + 45(-\frac{1}{6})$ is the statistics-*blind* plain trace $\text{Tr}[k_1]$ over the SM tower (4 minimal scalars, 12 massless vectors, 45 Weyl fermions). The signed, spin-weighted supertrace — bosons *minus* fermions, Visser’s Eq. 12 — evaluates on the same tower, with per-species weights verified against the primary sources (Visser gr-qc/0204062 Table 1: scalar $+\frac{1}{6} - \xi$, Weyl $-\frac{1}{6}$, Dirac $-\frac{1}{3}$, massless vector $-\frac{2}{3}$; Larsen–Wilczek hep-th/9506066 Secs. 3.4–3.5: each spin- $\frac{1}{2}$ d.o.f. enters with the *same* sign as a scalar), to $\text{str}[k_1]_{\text{SM}} = 4(\frac{1}{6} - \xi_H) + 12(-\frac{2}{3}) - 45(-\frac{1}{6}) = +\frac{1}{6} - 4\xi_H$ (48 Weyl, i.e. 3 right-handed ν : $+\frac{2}{3} - 4\xi_H$) — a 49-vs-48 near-cancellation in units of $\frac{1}{6}$ at minimal coupling, *sign-indefinite* in the un-pinned ξ_H (zero at $\xi_H = \frac{1}{24}$, resp. $\frac{1}{6}$). Under the anchor map confirmed by the

printed pair, $g = 2\pi/|\text{str}[k_1]|$, the corrected value gives $g = 37.7$ (9.42) and $E_\star = 6.14$ (3.07) E_{Pl} — super-Planckian. Accordingly the parent (run-112 restatement) no longer derives the anchor from an SM evaluation: H*2 is consumed as a *posited* $\mathcal{O}(1)$ input — sign and magnitude content- and ξ_H -dependent, the 0.42/0.65 headline and the definite-sign clause surrendered, $E_\star = \mathcal{O}(1\text{--}6) E_{\text{Pl}}$ priced against the (H0) clause $E_\star \lesssim E_{\text{Pl}}$ and the CHL window $(2\text{--}4) E_{\text{Pl}}$. Lane 2 is re-derived against that posited anchor; nothing below consumes -14.83 . Two facts fix the accounting. (i) $\text{str}[k_1]$ is *mass-independent* to leading order: it is the coefficient of the quadratically divergent $\propto E_\star^2$ Sakharov piece, a democratic signed spin-weight count of every sub-cutoff field, with mass entering only at $\mathcal{O}(m^2/E_\star^2)$. A new field’s shift $\Delta\text{str}[k_1]$ is its net signed spin-weight, mass-free. (ii) *Withdrawn and reversed (run 112)*. This item previously read: “a single $\mathcal{O}(1)$ shift ($\sim 7\%$ of $|\text{str}[k_1]|$) is *inside* tolerance” — arithmetic against the superseded $|\text{str}[k_1]| \approx 14.83$, on which a dark vector was 4.5%. Against the corrected $|\text{str}[k_1]|_{\text{SM}} = \frac{1}{6}$ the same shifts are 100–400% of the *entire* SM value: one minimal scalar or Weyl fermion ($\Delta\text{str} = +\frac{1}{6}$) is 100%, one Dirac fermion ($+\frac{1}{3}$) 200%, one massless dark vector ($-\frac{2}{3}$) 400% — the latter flips the 45-Weyl total ($+\frac{1}{6} \rightarrow -\frac{1}{2}$) and, on the 48-Weyl variant at $\xi_H = 0$, cancels it *exactly*. No candidate is “inside tolerance”: every individual field re-prices the anchor by an $\mathcal{O}(1)$ factor or can flip or null it. What survives is posited-anchor-relative and weaker: any single candidate leaves $|\text{str}[k_1]| \leq \mathcal{O}(1)$, so the posit is not violated — but the anchor’s value, sign, and $E_\star$ cannot be quoted independently of dark-sector content (and are ξ_H -indefinite in-SM).

The separate running-vacuum coefficient *is* mass-weighted, $\nu_{\text{eff}} = \frac{1}{2\pi}(\frac{1}{6}-\xi)M_X^2/E_{\text{Pl}}^2[1+\mathcal{O}(m^2/M_X^2)]$ with M_X the heaviest contributing field; a Planck-mass field gives $\nu \sim \frac{1}{2\pi} \cdot \frac{1}{6} \approx 3 \times 10^{-2}$, and a field at the corrected posited $E_\star = \mathcal{O}(1\text{--}6) E_{\text{Pl}}$ a fortiori more (the figure previously printed here, $\nu \sim 10^{-2}$ at “the native $E_\star$ ”, consumed the superseded $E_\star = 0.65 E_{\text{Pl}}$) — excluded by more than an order of magnitude by $\dot{G}/G$ and running-vacuum bounds, while every sub-Planckian candidate gives $\nu_{\text{eff}} = (m/E_{\text{Pl}})^2 \lll$ the bound. This mass-weighted leg is otherwise *untouched* by the run-112 statistics correction.

Candidate class	scale	$\Delta\text{str}[k_1]$	$\nu_{\text{eff}} = (m/E_{\text{Pl}})^2$	Verdict (vs. $ \text{str} _{\text{SM}=\frac{1}{6}, \xi_H=0}$)
(a) WIMP-class field	GeV–TeV	$+\frac{1}{6} / +\frac{1}{3}$ (Maj./Dirac)	$6.7 \times 10^{-33} \dots 10^{-39}$	TOLERATED; re-prices anchor by 100–200%
(b) ultralight/fuzzy scalar	10^{-22} eV	$+(\frac{1}{6} - \xi)$ (0 if conformal)	6.7×10^{-101}	TOLERATED; 0–100%
(c) dark photon (massless U(1))	$m = 0$	$-\frac{2}{3}$ exactly	0	TOLERATED; 400%: flips (45W) or <i>nulls</i> (48W) the total
(d) sterile- ν -class fermion	keV	$+\frac{1}{6}$ per Weyl	6.7×10^{-51}	TOLERATED; 100% each (three = 48W variant)
(e) PBH (no new field)	—	0	0	TOLERATED (trivially; unchanged)
(f) full hidden sector	many fields	$\sum$ net signed	(heaviest field)	CONSTRAINED (conditional band); EXCL. clause withdrawn

The honest inversion (run 112; replaces the previous “honest null”). The withdrawn text asserted insensitivity — single fields shift $\text{str}[k_1]$ “inside the $\mathcal{O}(1)$ tolerance”; only a hidden

sector with net aligned spin-weight $\gtrsim |\text{str}[k_1]| \sim 15$ was **CONSTRAINED**; net positive weight $\gtrsim 15$ (“flipping $\text{str}[k_1]$ ’s sign, $g < 0$ ”) was **EXCLUDED**. All three clauses were arithmetic against the superseded plain-trace -14.83 and are withdrawn. (1) *No insensitivity*: against $|\text{str}[k_1]|_{\text{SM}} = \frac{1}{6}$ every single candidate class shifts the supertrace by 100–400% (table); the anchor is co-determined by the dark sector, not a constraint on it. Any single field still leaves $|\text{str}[k_1]| \leq \mathcal{O}(1)$, so the *posited* anchor survives, but $E_\star$ is re-priced per field ($6.14 \rightarrow 4.34 E_{\text{Pl}}$ for one minimal scalar, $6.14 \rightarrow 3.54 E_{\text{Pl}}$ for one Dirac; 45-Weyl, $\xi_H = 0$). (2) *No sign-flip exclusion*: the corrected baseline is itself sign-indefinite in ξ_H (zero at $\frac{1}{24}$; a conformally coupled Higgs alone gives $-\frac{1}{2}$) and the restated H*2 surrenders the definite-sign clause; a criterion excluding candidates for flipping a sign the SM does not fix is empty — the **EXCLUDED** verdict is withdrawn, not re-graded, and nothing replaces it. (3) *What (f) retains — a conditional band*: priced against the CHL window $E_\star \in (2-4) E_{\text{Pl}}$, the posited anchor needs total $|\text{str}[k_1]| \in (2\pi/16, 2\pi/4) \approx (0.39, 1.57)$: net aligned dark weight $\sim 1-9$ units of $\frac{1}{6}$ beyond the SM’s $+\frac{1}{6}$ (positive branch; $\sim 3-10$ units, $\approx 1-2.5$ vectors, negative) — a two-sided band, not an upper bound. On the strict (H0) reading $E_\star \lesssim E_{\text{Pl}}$ the inversion completes: $|\text{str}[k_1]| \geq 2\pi$ would *require* net aligned weight $\gtrsim 37$ units (≈ 37 scalar-equivalents or $\gtrsim 10$ dark vectors) — the configuration the withdrawn text **EXCLUDED** is what that reading would demand. The parent prices this tension rather than resolving it; we record the arithmetic and make no field-content claim (the identity fence binds). *Grade*: **PLAUSIBLE**, re-scoped — the corrected $\text{str}[k_1]_{\text{SM}} = +\frac{1}{6} - 4\xi_H$ ($+\frac{2}{3} - 4\xi_H$ with right-handed neutrinos) rests on source-verified weights (Visser gr-qc/0204062 Table 1; Larsen–Wilczek hep-th/9506066) and exact row arithmetic; the superseded -14.83 was the statistics-blind $\text{Tr}[k_1]$, its billing as “framework-internal SM evaluation” revoked; the (f) band is conditional on the parent’s *posited* H*2, an assumed $\mathcal{O}(1)$ input, not a derived value.

4 Lane 3: the machinery taxonomy (TOUCHES)

4.1 Massless dark U(1) vs. the $m = 0$ IR gate

A massless dark U(1) — pure dark Maxwell or dark QED with fermionic matter, *no* dark scalar — does *not* carry the b_0 improvement site and does *not* inherit the $m = 0$ IR-gate wall. Its physical degrees of freedom are transverse (co-exact 1-form, b_1 sector), closed by the geometric floor ω_1 ; it has no gauge-invariant dimension-2 scalar, hence no $c' : \varphi^2$: improvement seed (free-Maxwell $T^\mu_\mu = 0$), and its gauge-fixing quartet is BRST-exact and emptied by Kugo–Ojima. The b_0 wall is *scalar-improvement-generic*, not gauge-sector-generic — the “gauge-sector generic” reading is refuted, since the gauge sector is precisely the one that closes. A Higgsed dark U(1) carries the site via its dark *scalar* (§4.2), never via the gauge boson. *Grade*: **THEOREM** (*taxonomy/applicability*).

4.2 Ultralight scalar DM vs. the QEI floors

A fuzzy scalar at $m \sim 10^{-22}$ eV has $m > 0$ — it is not strictly massless — so the free-scalar QEI floor $\langle \hat{T}^{\text{ren}}_{\mu\nu} [t^\mu t^\nu F^2] \rangle_\omega \geq -c_S(g, t, F)$ applies to it *directly*, and it is the *best*-served candidate class, not the worst. The floor does *not* degrade in the ultralight limit: on the load-bearing one-sided floor c_S the mass enters only as *forward* (tightening) powers — $m > 0$ strictly tightens it, and the *massless* limit is finite (the massless Dirac floor is $\frac{4}{3} \times$ the scalar one) — so as m shrinks the floor relaxes *monotonically toward its finite massless value*. The only inverse-mass factors — the upper-bound constant’s $1/m^2$ and the modular-trace $1/m$ — sit on sides the chain consumes nowhere, are decoupled from the floor, are finite for any $m > 0$, and are geometrically re-anchored to the lowest seed frequency ω_1 . The controlling IR scale is therefore slice *geometry*, not the candidate

mass; the only honest wall is gapless horizon spectra ($\omega_1 \rightarrow 0$), a mass-independent slice statement lying $\gtrsim 11$ orders below any dark-matter mass scale. *Grade: THEOREM* (floor non-degradation, print + certified); TOUCHES (an applicability statement, not a constraint on m).

4.3 PBH dark matter vs. the horizon machinery

For primordial black holes the framework introduces no new field (§3, row (e), $\Delta\text{str}[k_1] = 0$). The horizon machinery applies *per object*, split by the object’s dynamical state:

- *Stationary* (the asteroid-mass window $\sim 10^{17}\text{--}10^{23}$ g — the only window in which PBHs can be all of the dark matter, and near-stationary because the evaporation time far exceeds a Hubble time): the type-II / modified first law $T_H \delta S = \delta\langle\hat{H}_c\rangle + \alpha \delta\langle\Sigma_\delta t\rangle$, the free first law of entanglement, and the generalised entropy $S_{\text{gen}} = A/4G + S_{\text{bulk}}$ hold at each PBH horizon at its own κ , T_H , A . *THEOREM*, mass-independent in applicability.
- *Evaporating* (the endpoint / non-DM regime): the Type II_∞ structure, unbounded S_{gen} , and the Page curve hold, *conditional* on the ambient quasi-local modular-flow hypothesis. *CONDITIONAL THEOREM*; this conditionality attaches to the non-DM branch, not to the viable-DM PBHs.

These are statements of *applicability*, not constraints on PBH-as-DM, and assert nothing about dark-matter identity.

4.4 DM decoherence channel (kept tight)

The framework’s conditioned-source/instrument structure and its horizon-decoherence floor describe the decoherence of *quantum* sources. Whether they imply any dark-matter-specific decoherence channel cannot be derived without asserting dark-sector field content, which the identity fence forbids. [SPECULATION.] No such channel is claimed. (*Stop.*)

5 Summary: requires / constrains / touches

REQUIRES	genuine $\langle T_{\mu\nu} \rangle$ for all galactic-scale (rotation-curve / lensing / cluster) phenomenology; no gravitational mimic exists (Prop. 1).
CONSTRAINS	(run-112 correction) <i>nothing outright</i> : with $\text{str}[k_1]_{\text{SM}} = +\frac{1}{6} - 4\xi_H$ — not the superseded -14.83 — and the anchor a posited $\mathcal{O}(1)$ input, no candidate class is excluded; the old $N_{\text{dark}} \lesssim \mathcal{O}(10\text{--}100)$ bound and the sign-flip exclusion are withdrawn. A hidden sector’s net aligned weight is <i>banded</i> conditionally on the posited anchor (§3); every single candidate re-prices the anchor at $\mathcal{O}(1)$; and still <i>no</i> candidate is constrained via $\nu_{\text{eff}}/\text{CC}$.
TOUCHES	the $m = 0$ gate does <i>not</i> touch a pure/fermionic dark U(1) (scalar-generic); the QEI floors touch ultralight scalars (finite, non-degrading); the type-II / Bekenstein first laws touch every PBH horizon (availability).

The honest ceiling. The framework REQUIRES genuine stress-energy for the dark sector; after the run-112 correction its anchor accounting CONSTRAINS no candidate outright — it conditionally bands aggregate spin-weight under a *posited* $\mathcal{O}(1)$ anchor whose value the dark sector co-determines — and its theorems TOUCHES candidate classes in the taxonomy above. It *does not, and by construction cannot*, say what dark matter is.